

THE $3x + 1$ CONJUGACY MAP SENDS THE CRITICAL STURMIAN WORD TO AN IRRATIONAL 2-ADIC INTEGER

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ABSTRACT. Let T be the $3x + 1$ map on the 2-adic integers and let Φ be the Bernstein–Lagarias conjugacy map, which sends a parity vector to the unique 2-adic integer realising it. Let $\beta = \ln 2 / \ln 3$ and let $1c_\beta$ be the Sturmian word of slope β studied by López and Stoll, the parity word of critical ones-density. We prove that $\Phi(1c_\beta)$ is irrational, that is, its 2-adic expansion is not eventually periodic. The proof is elementary and effective: the periodic words attached to the convergents p_n/q_n of β have Φ -values $c_n/(2^{q_n} - 3^{p_n})$ whose numerators are small because the words are mechanical, while at the convergents with $p_n/q_n > \beta$ these values agree with $\Phi(1c_\beta)$ to 2-adic depth exactly $q_n + q_{n+1} - 1$; a Liouville comparison between the two closes the argument, using no property of the continued fraction of β beyond $q_n \rightarrow \infty$. Consequences: the same holds for $\Phi(c_\beta)$, for $\Phi(0c_\beta)$ and for every shift of $1c_\beta$; and for no odd $q \geq 1$ does the generalised $3x + q$ map possess an integer orbit whose parity word is $1c_\beta$. The periodic approximants and their approximation depths go back to López and Stoll (2009), who state the corresponding 2-adic question as open; no prior proof is known.

1. INTRODUCTION

Let $T: \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ be the $3x + 1$ map in its standard form,

$$T(x) = \begin{cases} x/2, & x \equiv 0 \pmod{2}, \\ (3x + 1)/2, & x \equiv 1 \pmod{2}. \end{cases}$$

The *parity vector* of $x \in \mathbb{Z}_2$ is the sequence $v(x) = (v_0, v_1, \dots) \in \{0, 1\}^{\mathbb{N}}$ with $v_i \equiv T^i(x) \pmod{2}$. Bernstein and Lagarias [2, 1] showed that $x \mapsto v(x)$ is a bijection onto $\{0, 1\}^{\mathbb{N}}$, indeed a 2-adic isometry once $\{0, 1\}^{\mathbb{N}}$ is identified with $\mathbb{Z}_2$ through binary expansions; we write Φ for its inverse, so that $\Phi(v)$ is the unique 2-adic integer whose parity vector is v . Conjugating T by Φ turns it into the shift.

Rationality of $\Phi(v)$ corresponds to eventual periodicity of v in one direction only. If v is eventually periodic then $\Phi(v) \in \mathbb{Q}$; the converse is Lagarias’s Periodicity Conjecture [6, §2], which is open, and which asserts that no aperiodic parity vector has a rational Φ -value. Very little is known unconditionally in that direction. López and Stoll [7] studied Φ on Sturmian parity words and produced a 2-adic series and a generalised continued fraction for the values. They state in their abstract that it is unknown whether there exists any aperiodic v with an eventually periodic $\Phi(v)$, and that their computed examples suggest that Φ always maps Sturmian words to words of full complexity.

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This paper settles one explicit word. Let

$$\beta = \frac{\ln 2}{\ln 3} = \log_3 2 = 0.63092975\dots, \quad \alpha = 1/\beta = \log_2 3,$$

and let $1c_\beta$ be the Sturmian word of slope β and intercept 0 in the notation of [7], namely $1c_\beta(j) = \lceil (j+1)\beta \rceil - \lceil j\beta \rceil$ for $j \geq 0$. Its ones-density is β , the critical density for the $3x+1$ map: by (1) below, an orbit whose parity word has $k_\ell/\ell \rightarrow \beta$ satisfies $(3^{k_\ell}/2^\ell)^{1/\ell} \rightarrow 3^\beta/2 = 1$, so its multiplicative growth rate is exactly 1.

Theorem 1.1. $\Phi(1c_\beta) \notin \mathbb{Q}$. *Equivalently, the 2-adic expansion of $\Phi(1c_\beta)$ is not eventually periodic.*

The critical density is exactly where the existing methods stop. López and Stoll's irrationality results in [7] concern the *real* function $\Phi_{\mathbb{R}}$ and hold on $\mathbb{Q} \cap (\beta, 1]$, with a dual construction below β ; the restriction is forced, because the relevant real series converges if and only if $\beta < p/q \leq 1$, and their staircase diverges at $x = \beta$. Their later preprint [8] proves that Φ maps an aperiodic parity word whose *lower* ones-density is *strictly* greater than β to an aperiodic 2-adic integer, and its Theorem 1 shows that a divergent trajectory of a rational 2-adic integer would have lower ones-density exactly β . The word of Theorem 1.1 sits at that single excluded density, and the present argument is 2-adic rather than archimedean.

Method. Three ingredients, each elementary.

- (i) For a finite word w of length ℓ with k ones, the periodic word w^∞ has $\Phi(w^\infty) = c_w/(2^\ell - 3^k)$ with c_w an explicit positive integer (Proposition 4.1). When w is *mechanical*, $c_w \leq \ell \max(2^\ell, 3^k)$ (Lemma 4.2); for general w no such bound holds, at any constant (Remark 4.3).
- (ii) Let p_n/q_n be the convergents of β and w_n the mechanical word of slope p_n/q_n . Then $\Phi(w_n^\infty)$ agrees with $\Phi(1c_\beta)$ to 2-adic depth exactly $q_n - 1$ when $p_n/q_n < \beta$, and to depth exactly $q_n + q_{n+1} - 1$ when $p_n/q_n > \beta$ (Theorem 5.3). The asymmetry is the point: $1c_\beta$ is approached from one side, and the convergents on that side see one full continued-fraction level further.
- (iii) If $\Phi(1c_\beta) = u/v$ then the integer $M_n = u\delta_n - v c_n$ is nonzero with $v_2(M_n)$ equal to the depth in (ii), so $|M_n| \geq 2^{q_n + q_{n+1} - 1}$, while (i) gives $|M_n| \leq 4Hq_n 2^{q_n}$ with $H = \max(|u|, v)$. Hence $q_{n+1} \leq 3 + \log_2 H + \log_2 q_n$, which fails for large n (Theorem 6.1).

No property of the continued fraction of β is used beyond $q_n \rightarrow \infty$: no irrationality measure for β , no bound on its partial quotients. The argument is effective, and yields an explicit height below which no rational can lie (Corollary 6.2).

Credit. The periodic approximants and their approximation depths are not new. For every irrational slope, [7, Theorem 1] gives a closed 2-adic series for $\Phi(1c_\beta)$ indexed by consecutive pairs of continued-fraction convergents, whose term exponents are $q_{j+1} + q_j - 1$; these are the depths of Theorem 5.3 at the convergents from above. A companion technical note of the author [4] derived the depths at both parities independently, and more sharply, in the coordinates of the accelerated $3x+1$ map. What is new here is ingredient (iii), the Liouville comparison, and the irrationality conclusion. No prior proof is known; the question is stated as open in López–Stoll (2009) and treated as open in López–Stoll (2021).

Scope. Theorem 1.1 concerns one explicit word of critical density, together with its shifts and the two variants $c_\beta, 0c_\beta$. It says nothing about divergent orbits of positive integers, nothing about the Periodicity Conjecture in general — which quantifies over all aperiodic parity vectors,

an uncountable family — and nothing about the Collatz conjecture. The paper is self-contained: every object is defined here, and no machinery from [4, 5] is needed to read it.

2. THE CONJUGACY MAP

Throughout, v_2 is the 2-adic valuation on $\mathbb{Z}_2$ with $v_2(0) = \infty$, and for $v, v' \in \{0, 1\}^{\mathbb{N}}$ we write $\text{lcp}(v, v')$ for the length of their longest common prefix. For $v \in \{0, 1\}^{\mathbb{N}}$ and $m \geq 0$ put

$$k_m(v) = \#\{i < m : v_i = 1\},$$

the number of ones among $v_0, \dots, v_{m-1}$; thus $k_0(v) = 0$.

The one-step identity $T(x) = (3^{v_0}x + v_0)/2$, valid for both parities, gives by induction the standard formula

$$T^n(x) = \frac{3^{k_n}x + c_n}{2^n}, \quad c_n = c_n(v) = \sum_{i < n, v_i=1} 3^{k_n - k_{i+1}} 2^i, \quad (1)$$

where $v = v(x)$ and $k_m = k_m(v)$. Indeed $c_0 = 0$ and $c_{n+1} = 3^{v_n}c_n + v_n 2^n$.

Proposition 2.1. *Let $v \in \{0, 1\}^{\mathbb{N}}$ contain infinitely many ones. Then the series below converges in $\mathbb{Z}_2$ and*

$$\Phi(v) = - \sum_{i: v_i=1} 3^{-k_{i+1}(v)} 2^i.$$

Proof. Each $3^{-k_{i+1}}$ is a 2-adic unit and $v_2(3^{-k_{i+1}} 2^i) = i$, so the terms tend to 0 and the series converges. Let $x = \Phi(v)$. Rearranging (1) gives $3^{k_n}x = 2^n T^n(x) - c_n$, whence

$$x = \frac{2^n T^n(x)}{3^{k_n}} - \frac{c_n}{3^{k_n}} = \frac{2^n T^n(x)}{3^{k_n}} - \sum_{i < n, v_i=1} 3^{-k_{i+1}} 2^i.$$

Since $T^n(x) \in \mathbb{Z}_2$ and 3^{-k_n} is a unit, the first term has valuation at least n , so it tends to 0; letting $n \rightarrow \infty$ gives the claim. $\square$

Proposition 2.2. *For all $v, v' \in \{0, 1\}^{\mathbb{N}}$,*

$$v_2(\Phi(v) - \Phi(v')) = \text{lcp}(v, v').$$

Proof. Fix $L \geq 1$. By (1) with $n = L$ we have $3^{k_L}x \equiv -c_L \pmod{2^L}$, so $x \equiv -c_L 3^{-k_L} \pmod{2^L}$; and k_L, c_L depend only on $v_0, \dots, v_{L-1}$. Hence $x \pmod{2^L}$ is determined by the first L letters of $v(x)$. Conversely $v_0, \dots, v_{L-1}$ are determined by $x \pmod{2^L}$, by induction on L : $v_0 \equiv x \pmod{2}$, and $T(x) \pmod{2^{L-1}}$ is determined by $x \pmod{2^L}$. So $v \mapsto \Phi(v) \pmod{2^L}$ is a bijection from $\{0, 1\}^L$ onto $\mathbb{Z}/2^L\mathbb{Z}$. Consequently $\Phi(v) \equiv \Phi(v') \pmod{2^L}$ if and only if v and v' agree in their first L letters, which is the assertion. $\square$

Proposition 2.3. *Let σ denote the shift on $\{0, 1\}^{\mathbb{N}}$. Then $\Phi(\sigma v) = T(\Phi(v))$, and $\Phi(v)$ is odd if and only if $v_0 = 1$.*

Proof. The parity vector of $T(x)$ is $\sigma v(x)$ by definition, and $v_0(x) \equiv x \pmod{2}$. $\square$

3. THE CRITICAL STURMIAN WORD

Let $\beta = \log_3 2 \in (0, 1)$ and $\alpha = 1/\beta = \log_2 3$, both irrational. In the notation of [7],

$$1c_\beta(j) = \lceil (j+1)\beta \rceil - \lceil j\beta \rceil, \quad 0c_\beta(j) = \lfloor (j+1)\beta \rfloor - \lfloor j\beta \rfloor \quad (j \geq 0),$$

and c_β denotes the characteristic word $c_\beta(j) = \lfloor (j+1)\beta \rfloor - \lfloor j\beta \rfloor$ for $j \geq 1$. Since $\beta \in (0, 1)$ is irrational, $1c_\beta = 1c_\beta$ and $0c_\beta = 0c_\beta$.

Lemma 3.1. *For $n \geq 0$ we have $k_n(1c_\beta) = \lceil n\beta \rceil$, and for $j \geq 0$ the $(j+1)$ -st one of $1c_\beta$ occupies position $\lfloor j\alpha \rfloor$.*

Proof. The first claim is a telescoping sum. For the second, position i carries the $(j+1)$ -st one exactly when $k_i(1c_\beta) = j$ and $k_{i+1}(1c_\beta) = j+1$, that is, $\lceil i\beta \rceil \leq j$ and $\lceil (i+1)\beta \rceil \geq j+1$. For $j \geq 1$ the first reads $i\beta \leq j$, i.e. $i \leq j\alpha$, i.e. $i \leq \lfloor j\alpha \rfloor$ since $j\alpha \notin \mathbb{Z}$; the second reads $(i+1)\beta > j$, i.e. $i+1 > j\alpha$, i.e. $i \geq \lfloor j\alpha \rfloor$. Hence $i = \lfloor j\alpha \rfloor$. For $j = 0$ both conditions force $i = 0 = \lfloor 0 \cdot \alpha \rfloor$. $\square$

Proposition 3.2. *Let $\Xi_\alpha = \sum_{j \geq 0} 3^{-(j+1)} 2^{\lfloor j\alpha \rfloor} \in \mathbb{Z}_2$. Then $\Phi(1c_\beta) = -\Xi_\alpha$. In particular $\Phi(1c_\beta)$ is a 2-adic unit.*

Proof. By Lemma 3.1 the ones of $1c_\beta$ are at the positions $\lfloor j\alpha \rfloor$, $j \geq 0$, and the one at position $\lfloor j\alpha \rfloor$ is the $(j+1)$ -st, so $k_{i+1} = j+1$ there. Proposition 2.1 gives the formula. The term $j = 0$ has valuation 0 and all others have positive valuation, so $v_2(\Xi_\alpha) = 0$. $\square$

The constant Ξ_α is the 2-adic Sturmian carry constant of [4]; Proposition 3.2 is the dictionary between the two normalisations, and we shall not need it again. All statements below are about Φ .

4. PERIODIC VALUES AND THEIR HEIGHTS

For a finite word $w = w_0 \cdots w_{\ell-1} \in \{0, 1\}^\ell$ with $k \geq 1$ ones we write w^∞ for its infinite periodic repetition and put

$$c_w = \sum_{i < \ell, w_i = 1} 3^{k-k_{i+1}(w)} 2^i \in \mathbb{Z}_{>0}. \quad (2)$$

Proposition 4.1. *With the notation above, $2^\ell \neq 3^k$ and*

$$\Phi(w^\infty) = \frac{c_w}{2^\ell - 3^k}.$$

Proof. $2^\ell = 3^k$ is impossible for $k \geq 1$ by unique factorisation. Write $i = t\ell + r$ with $0 \leq r < \ell$; periodicity gives $k_{i+1}(w^\infty) = tk + k_{r+1}(w)$. Grouping the sum of Proposition 2.1 by periods,

$$\Phi(w^\infty) = - \sum_{t \geq 0} \sum_{r < \ell, w_r = 1} 3^{-(tk+k_{r+1}(w))} 2^{t\ell+r} = - \left(\sum_{r < \ell, w_r = 1} 3^{-k_{r+1}(w)} 2^r \right) \sum_{t \geq 0} (3^{-k} 2^\ell)^t.$$

The geometric series converges in $\mathbb{Z}_2$ because $v_2(3^{-k} 2^\ell) = \ell \geq 1$, with sum $(1 - 3^{-k} 2^\ell)^{-1} = 3^k / (3^k - 2^\ell)$. Since $c_w = 3^k \sum_{r < \ell, w_r = 1} 3^{-k_{r+1}(w)} 2^r$, multiplying gives $\Phi(w^\infty) = -c_w / (3^k - 2^\ell)$, which is the stated value. $\square$

For coprime $1 \leq p \leq q$ let $w_{p,q}$ be the *mechanical word of slope p/q* : the word of length q whose ones occupy the positions $\lfloor jq/p \rfloor$ for $0 \leq j < p$. It has exactly p ones.

Lemma 4.2. *Let $w = w_{p,q}$. Then $0 < c_w \leq q \cdot \max(2^q, 3^p)$.*

Proof. Positivity is clear from (2). For $0 \leq i < q$,

$$k_{i+1}(w) = \#\{0 \leq j < p : \lfloor jq/p \rfloor \leq i\} = \#\{0 \leq j < p : j < (i+1)p/q\} \geq (i+1)p/q,$$

the last step because the count is $\min(p, \lceil (i+1)p/q \rceil)$ and $(i+1)p/q \leq p$. Hence $p - k_{i+1}(w) \leq p(q-1-i)/q$ and, with $\rho = 3^{p/q}$,

$$3^{p-k_{i+1}(w)} 2^i \leq \rho^{q-1-i} 2^i \leq \max(\rho^{q-1}, 2^{q-1}) \leq \max(3^p, 2^q),$$

where the middle inequality holds because $i \mapsto \rho^{q-1-i}2^i$ is monotone, so its maximum over $0 \leq i \leq q-1$ is attained at an endpoint. The sum (2) has at most q terms. $\square$

Remark 4.3. The hypothesis that w be mechanical cannot be dropped, and not merely the constant is at stake. Take $w = 0^a 1^a$, so $\ell = 2a$ and $k = a$. Its ones are at $i = a, \dots, 2a-1$ with $k_{i+1}(w) = i - a + 1$, so by (2)

$$c_w = \sum_{t=0}^{a-1} 3^{a-1-t} 2^{a+t} = 2^a \sum_{t=0}^{a-1} 3^{a-1-t} 2^t = 2^a (3^a - 2^a).$$

Here $\max(2^\ell, 3^k) = 4^a$, so

$$\frac{c_w}{\ell \cdot \max(2^\ell, 3^k)} = \frac{3^a - 2^a}{2a \cdot 2^a} = \frac{(3/2)^a - 1}{2a} \xrightarrow{a \rightarrow \infty} \infty.$$

Thus no bound $c_w \leq C\ell \max(2^\ell, 3^k)$ holds for arbitrary w , for any constant C . Numerically the ratio is 2.83 at $a = 10$, 14.56 at $a = 15$ and 41.03 at $a = 18$. For an arbitrary *balanced* w the proof of Lemma 4.2 goes through with the discrepancy bound $|k_{i+1}(w) - (i+1)k/\ell| < 1$ in place of the exact count, at the cost of a factor 3.

5. THE APPROXIMATION DEPTH

Convention 5.1. p_n/q_n denotes the n -th convergent of the continued fraction

$$\beta = [0; 1, 1, 1, 2, 2, 3, 1, 5, 2, 23, \dots],$$

indexed from $p_0/q_0 = 0/1$, so that $p_1/q_1 = 1/1$ and $p_2/q_2 = 1/2$. Thus p_n counts ones and q_n counts letters. We abbreviate $w_n = w_{p_n, q_n}$ and $\alpha_n = q_n/p_n$, and write $D_n = |q_n\beta - p_n|$, so that the classical estimate $D_n < 1/q_{n+1}$ holds.

The first convergents are

n	1	2	3	4	5	6	7
p_n/q_n	1/1	1/2	2/3	5/8	12/19	41/65	53/84
n	8	9	10	11	12	13	14
p_n/q_n	$\frac{306}{485}$	$\frac{665}{1054}$	$\frac{15601}{24727}$	$\frac{31867}{50508}$	$\frac{79335}{125743}$	$\frac{111202}{176251}$	$\frac{190537}{301994}$

Because the convergents alternate about β and $p_1/q_1 = 1 > \beta$, we have $p_n/q_n > \beta$ exactly when n is odd. Equivalently, comparing exactly, $3^{p_n} > 2^{q_n}$ if and only if n is odd. Also $q_{n+1} > q_n$ for $n \geq 1$.

Lemma 5.2 (floor comparison). *Let $n \geq 3$ and $j_n^* = \min\{j \geq 1 : \lfloor j\alpha \rfloor \neq \lfloor j\alpha_n \rfloor\}$.*

- (1) *If n is odd, then $j_n^* = p_n + p_{n+1}$, and at that index $\lfloor j_n^*\alpha \rfloor = q_n + q_{n+1}$ while $\lfloor j_n^*\alpha_n \rfloor = q_n + q_{n+1} - 1$.*
- (2) *If n is even, then $j_n^* = p_n$, and at that index $\lfloor j_n^*\alpha \rfloor = q_n - 1$ while $\lfloor j_n^*\alpha_n \rfloor = q_n$.*

Proof. Write $\alpha_n - \alpha = \eta_n$. Since $\alpha_n = q_n/p_n$ and $\alpha = 1/\beta$,

$$\eta_n = \frac{q_n}{p_n} - \frac{1}{\beta} = \frac{q_n\beta - p_n}{p_n\beta},$$

so $\eta_n < 0$ when n is odd and $\eta_n > 0$ when n is even, and in both cases $|\eta_n| = D_n/(p_n\beta)$.

Set $m_j = jq_n \bmod p_n$, so that $\{j\alpha_n\} = m_j/p_n$; since $\gcd(p_n, q_n) = 1$ we have $m_j \geq 1$ for $1 \leq j < p_n$, and $m_{p_n} = 0$.

(2), n even. Over the range $j \leq p_n$ used here the two floor sequences differ by at most one, since $j|\eta_n| \leq p_n|\eta_n| = \alpha D_n < \alpha/q_{n+1} < 1$; the equivalence below is therefore valid. Here $\eta_n > 0$, so a mismatch at j means $\lfloor j\alpha_n \rfloor > \lfloor j\alpha \rfloor$, which requires $\{j\alpha_n\} < j\eta_n$, i.e.

$m_j < jp_n\eta_n = jD_n/\beta = j\alpha D_n$. For $1 \leq j < p_n$ this fails, because $m_j \geq 1$ while $j\alpha D_n < p_n\alpha D_n < p_n\alpha/q_{n+1} \leq 1$, using $D_n < 1/q_{n+1}$ and $p_n\alpha < q_n < q_{n+1}$. At $j = p_n$ it holds, since $m_{p_n} = 0$. So $j_n^* = p_n$. The depths: $p_n\alpha_n = q_n$ exactly, so $\lfloor p_n\alpha_n \rfloor = q_n$; and $p_n\alpha = q_n - \alpha D_n$ with $0 < \alpha D_n < \alpha/q_{n+1} < 1$, so $\lfloor p_n\alpha \rfloor = q_n - 1$.

(1), n odd. Here $\eta_n < 0$, so a mismatch at j means $\lfloor j\alpha_n \rfloor < \lfloor j\alpha \rfloor$, which happens exactly when $\{j\alpha_n\} + j|\eta_n| \geq 1$, i.e.

$$m_j + j\alpha D_n \geq p_n. \quad (3)$$

We use two exact facts about consecutive convergents. First, $p_{n+1}q_n - p_nq_{n+1} = (-1)^n = -1$, so $p_{n+1}q_n \equiv -1 \pmod{p_n}$ and

$$m_{p_{n+1}} = p_n - 1, \quad p_{n+1}\alpha_n = q_{n+1} - \frac{1}{p_n}. \quad (4)$$

Second, writing $p_m = \beta q_m + (-1)^{m+1}D_m$ and using the classical identity $q_{n+1}D_n + q_nD_{n+1} = 1$,

$$(p_n + p_{n+1})D_n = \beta(q_n + q_{n+1})D_n + D_n(D_n - D_{n+1}) = \beta + (D_n - D_{n+1})(\beta q_n + D_n),$$

so, multiplying by $\alpha = 1/\beta$ and using $p_n = \beta q_n + D_n$,

$$(p_n + p_{n+1})\alpha D_n = 1 + p_n\alpha(D_n - D_{n+1}). \quad (5)$$

Note $0 < p_n\alpha(D_n - D_{n+1}) < (q_n + 1)D_n \leq (q_n + 1)/q_{n+1} \leq 1$, because $p_n\alpha = q_n + \alpha D_n < q_n + 1$ and $q_{n+1} \geq q_n + 1$; so the left side of (5) lies in $(1, 2)$. Subtracting $p_n\alpha D_n$ from (5) gives moreover

$$p_{n+1}\alpha D_n = 1 - p_n\alpha D_{n+1} < 1. \quad (6)$$

Finally, over the range of indices used below the two floor sequences differ by at most one: for $j \leq p_n + p_{n+1}$,

$$j|\eta_n| = \frac{j\alpha D_n}{p_n} \leq \frac{(p_n + p_{n+1})\alpha D_n}{p_n} < \frac{2}{p_n} \leq 1,$$

since $p_n \geq 2$ for $n \geq 3$. This is what makes (3) an equivalence.

Now no $j \leq p_{n+1}$ satisfies (3): since $m_j \leq p_n - 1$, it would force $j\alpha D_n \geq 1$, whereas $j\alpha D_n \leq p_{n+1}\alpha D_n < 1$ by (6). Next, for $j = p_{n+1} + i$ with $1 \leq i \leq p_n - 1$ we have $m_j = (iq_n \bmod p_n) - 1 \leq p_n - 2$ by (4), since $iq_n \not\equiv 0 \pmod{p_n}$; and $j\alpha D_n < (p_n + p_{n+1})\alpha D_n < 2$, so $m_j + j\alpha D_n < (p_n - 2) + 2 = p_n$ and (3) again fails. At $j^* = p_n + p_{n+1}$, however, $m_{j^*} = m_{p_{n+1}} = p_n - 1$ and, by (5),

$$m_{j^*} + j^*\alpha D_n = (p_n - 1) + 1 + p_n\alpha(D_n - D_{n+1}) = p_n + p_n\alpha(D_n - D_{n+1}) > p_n,$$

since $D_n > D_{n+1}$. Hence $j_n^* = p_n + p_{n+1}$.

The two depths are now exact. By (4), $j^*\alpha_n = q_n + p_{n+1}\alpha_n = q_n + q_{n+1} - 1/p_n$, and $p_n \geq 2$, so $\lfloor j^*\alpha_n \rfloor = q_n + q_{n+1} - 1$. Moreover $j^*\alpha = j^*\alpha_n + j^*\alpha D_n/p_n$, and by (5) $j^*\alpha D_n/p_n = 1/p_n + \alpha(D_n - D_{n+1})$, whence

$$j^*\alpha = q_n + q_{n+1} + \alpha(D_n - D_{n+1}), \quad 0 < \alpha(D_n - D_{n+1}) < \alpha D_n < \alpha/q_{n+1} < 1,$$

so $\lfloor j^*\alpha \rfloor = q_n + q_{n+1}$. □

Theorem 5.3 (depth law). *For every $n \geq 3$,*

$$v_2(\Phi(1c_\beta) - \Phi(w_n^\infty)) = \text{lcp}(1c_\beta, w_n^\infty) = \begin{cases} q_n - 1, & n \text{ even}, \\ q_n + q_{n+1} - 1, & n \text{ odd}. \end{cases}$$

Proof. The first equality is Proposition 2.2. For the second, both words are determined by their sets of one-positions: by Lemma 3.1 the ones of $1c_\beta$ are at $\lfloor j\alpha \rfloor$, $j \geq 0$, and by periodicity the ones of w_n^∞ are at $\lfloor j\alpha_n \rfloor$, $j \geq 0$ — indeed $\lfloor (j + p_n)\alpha_n \rfloor = \lfloor j\alpha_n \rfloor + q_n$. Both position sequences are strictly increasing, and they agree for $j < j_n^*$ by definition of j_n^* . Hence the two words agree at every position below $\min(\lfloor j_n^*\alpha \rfloor, \lfloor j_n^*\alpha_n \rfloor)$ and differ at that position, one of them carrying a

one there and the other a zero. So the longest common prefix has length $\min(\lfloor j_n^* \alpha \rfloor, \lfloor j_n^* \alpha_n \rfloor)$, which Lemma 5.2 evaluates as $q_n + q_{n+1} - 1$ for odd n and $q_n - 1$ for even n . $\square$

Remark 5.4. The two cases differ by a whole continued-fraction level, and only the odd case is of any use below: for even n the depth $q_n - 1$ is smaller than the archimedean size of the approximant, and the comparison of Section 6 is vacuous there. The reason for the asymmetry is that $1c_\beta$ is the limit of the mechanical words from one side, so a periodic extension taken from that side continues to agree with it for one level longer.

6. IRRATIONALITY

Theorem 6.1. $\Phi(1c_\beta) \notin \mathbb{Q}$. More precisely, suppose $\Phi(1c_\beta) = u/v$ with $u \in \mathbb{Z}$, $v \geq 1$, $\gcd(u, v) = 1$, and put $H = \max(|u|, v)$. Then every odd $n \geq 3$ satisfies

$$q_{n+1} \leq 3 + \log_2 H + \log_2 q_n. \quad (7)$$

Since $q_{n+1} > q_n$ and $q_n \rightarrow \infty$ along odd n , this is impossible; a contradiction has appeared by the first odd $n \geq 3$ with $q_n \geq 2 \log_2 H + 20$.

Proof. Assume $\Phi(1c_\beta) = u/v$ as stated. As $\Phi(1c_\beta) \in \mathbb{Z}_2$, the denominator v is odd. Fix an odd $n \geq 3$ and write

$$\delta_n = 2^{q_n} - 3^{p_n}, \quad c_n = c_{w_n}, \quad \Phi(w_n^\infty) = \frac{c_n}{\delta_n}$$

by Proposition 4.1. Since n is odd, $3^{p_n} > 2^{q_n}$, so $\delta_n < 0$ and $|\delta_n| = 3^{p_n} - 2^{q_n} < 3^{p_n} = \max(2^{q_n}, 3^{p_n})$. Put

$$M_n = u \delta_n - v c_n \in \mathbb{Z}, \quad \text{so} \quad \Phi(1c_\beta) - \Phi(w_n^\infty) = \frac{M_n}{v \delta_n}.$$

M_n is a nonzero integer of known valuation. Both v and δ_n are odd, so $v_2(M_n) = v_2(\Phi(1c_\beta) - \Phi(w_n^\infty))$, which Theorem 5.3 evaluates as $q_n + q_{n+1} - 1$. In particular the valuation is finite, so $M_n \neq 0$, and therefore

$$|M_n| \geq 2^{q_n + q_{n+1} - 1}. \quad (8)$$

Archimedean size. By Lemma 4.2, $c_n \leq q_n 3^{p_n}$. Hence

$$|M_n| \leq |u| |\delta_n| + v c_n \leq H \cdot 3^{p_n} + H \cdot q_n \cdot 3^{p_n} = H(1 + q_n) 3^{p_n} \leq 2Hq_n 3^{p_n}.$$

Now $3^{p_n} = 2^{p_n \alpha}$ and $p_n \alpha - q_n = \alpha(p_n - q_n \beta) = \alpha D_n$, with $0 < \alpha D_n < \alpha/q_{n+1} \leq \alpha/8 < 0.2$ because $n \geq 3$ forces $q_{n+1} \geq 8$. So $3^{p_n} < 2^{q_n + 0.2} < 1.15 \cdot 2^{q_n}$ and

$$|M_n| < 2.3 H q_n 2^{q_n} \leq 4Hq_n 2^{q_n}. \quad (9)$$

Comparison. Combining (8) and (9), $2^{q_n + q_{n+1} - 1} < 4Hq_n 2^{q_n}$, that is $2^{q_{n+1} - 1} < 4Hq_n$, and taking logarithms gives (7).

Finally, (7) together with $q_{n+1} > q_n$ gives

$$q_n - \log_2 q_n < 3 + \log_2 H. \quad (10)$$

Suppose $q_n \geq 2 \log_2 H + 20$ and put $Q = q_n$. Then $\log_2 H \leq (Q - 20)/2$, so the right-hand side of (10) is at most $3 + (Q - 20)/2 = Q/2 - 7$, while the left-hand side exceeds $Q/2 - 7$ because $Q/2 + 7 > \log_2 Q$ for every $Q \geq 1$. This contradicts (10). Since $t \mapsto t - \log_2 t$ is increasing for $t \geq 2$, every odd n with $q_n \geq 2 \log_2 H + 20$ is impossible; as there are infinitely many odd n and $q_n \rightarrow \infty$, such an n exists. $\square$

Corollary 6.2. $\Phi(1c_\beta)$ is not equal to any u/v in lowest terms with $\max(|u|, v) \leq 2^{301973}$.

Proof. Take $n = 13$, so $p_n/q_n = 111202/176251$ and $q_{n+1} = 301994$; the depth of Theorem 5.3 is $q_n + q_{n+1} - 1 = 478244$, a value independently confirmed by the exact computation of Appendix A. By (8) and (9), $2^{478244} \leq |M_{13}| < 4H \cdot 176251 \cdot 2^{176251}$, so $H > 2^{478244-176251}/705004 = 2^{301993}/705004 > 2^{301973}$, since $705004 < 2^{20}$. $\square$

Remark 6.3. The argument uses no property of the continued fraction of β beyond $q_n \rightarrow \infty$. In particular it needs no irrationality measure for β and no bound on its partial quotients, and it is effective: by Theorem 6.1 the contradiction appears at continued-fraction depth $O(\log \log H)$.

7. CONSEQUENCES

Corollary 7.1. $\Phi(c_\beta)$ and $\Phi(0c_\beta)$ are irrational.

Proof. By Proposition 2.3, $\Phi(c_\beta) = T(\Phi(1c_\beta))$ and $\Phi(c_\beta) = T(\Phi(0c_\beta))$, since $\sigma(1c_\beta) = \sigma(0c_\beta) = c_\beta$. As $1c_\beta$ begins with 1, the value $\Phi(1c_\beta)$ is odd and $\Phi(c_\beta) = (3\Phi(1c_\beta) + 1)/2$; as $0c_\beta$ begins with 0, the value $\Phi(0c_\beta)$ is even and $\Phi(c_\beta) = \Phi(0c_\beta)/2$. Both relations are $\mathbb{Q}$ -affine with rational coefficients and invertible, so the three values are rational or irrational together. Theorem 6.1 decides which. $\square$

Corollary 7.2. $\Phi(\sigma^k 1c_\beta)$ is irrational for every $k \geq 0$.

Proof. $\Phi(\sigma^k 1c_\beta) = T^k(\Phi(1c_\beta))$ by Proposition 2.3. It therefore suffices that $T(x) \in \mathbb{Q}$ if and only if $x \in \mathbb{Q}$. If $x \in \mathbb{Q} \cap \mathbb{Z}_2$ then $T(x) \in \mathbb{Q} \cap \mathbb{Z}_2$. Conversely, T is two-to-one on $\mathbb{Z}_2$: the preimages of y are $2y$, which is even, and $(2y - 1)/3$, which lies in $\mathbb{Z}_2$ because 3 is a unit and is odd; the parity of x selects the branch. Both expressions are rational when y is. $\square$

Corollary 7.3. Let $q \geq 1$ be odd and let T_q be the generalised map $T_q(n) = n/2$ for n even and $T_q(n) = (3n + q)/2$ for n odd. There is no $n_0 \in \mathbb{Z}$ whose T_q -orbit has parity word $1c_\beta$, nor any whose parity word is a shift $\sigma^k 1c_\beta$.

Proof. Suppose $n_0 \in \mathbb{Z}$ has T_q -orbit $(n_j)_{j \geq 0}$ with parity word w , that is, n_j is odd exactly when $w_j = 1$. Put $x_j = n_j/q \in \mathbb{Q}$. Then x_j is odd in $\mathbb{Z}_2$ exactly when n_j is odd, because q is odd, and

$$x_{j+1} = \frac{n_{j+1}}{q} = \frac{3n_j + w_j q}{2q} = \frac{3x_j + w_j}{2},$$

so $x_{j+1} = T(x_j)$ and the parity vector of x_0 is w . Hence $\Phi(w) = x_0 \in \mathbb{Q}$. Conversely, if $\Phi(w) = u/v$ in lowest terms with v odd, then $n_j = vT^j(\Phi(w))$ is an integer T_v -orbit with parity word w . The corollary is therefore equivalent to the irrationality of $\Phi(w)$, which is Theorem 6.1 for $w = 1c_\beta$ and Corollary 7.2 for its shifts. $\square$

8. WHY THE 2-ADIC PLACE

The series of Proposition 3.2 has no archimedean meaning at this slope, and this is not an accident of presentation. Writing $\lfloor j\alpha \rfloor = j\alpha - \{j\alpha\}$ and $2^\alpha = 3$,

$$3^{-(j+1)} 2^{\lfloor j\alpha \rfloor} = \frac{1}{3} 2^{-\{j\alpha\}} \in \left(\frac{1}{6}, \frac{1}{3}\right] \quad \text{for every } j \geq 0,$$

so the terms do not tend to 0 and the real series diverges. López and Stoll reach the same point from the other side: their real staircase diverges at $x = \beta$, and they call the object at that slope a divergent series.

The same identity $2^\alpha = 3$ places the evaluation point on the boundary of the domain of every relevant archimedean theorem. The Hecke–Mahler series $f(z_1, z_2) = \sum_{j \geq 0} z_1^j z_2^{\lfloor j\alpha \rfloor}$ is being evaluated at $(z_1, z_2) = (1/3, 2)$, where $|z_1 z_2^\alpha| = \frac{1}{3} \cdot 3 = 1$ exactly. Bugeaud and Laurent [3] prove transcendence at every algebraic point with $0 < |z_1|, |z_1 z_2^\alpha| < 1$ strictly, and give

explicit irrationality exponents at rational points strictly inside that domain; our point lies on its boundary. The results of Luca, Ouaknine and Worrell [10, 11] concern series $\sum_n f(\lfloor n\theta + \rho \rfloor) \gamma^{-n}$ with f a polynomial, so with coefficients of polynomial growth; here the coefficients $2^{\lfloor j\alpha \rfloor}$ grow like 3^j , which is precisely what makes the terms $\Theta(1)$.

At the 2-adic place the same resonance is harmless: $v_2(2^{\lfloor j\alpha \rfloor}) = \lfloor j\alpha \rfloor \rightarrow \infty$ while $3^{-(j+1)}$ is a unit, so the series converges, and the approximation problem becomes the comparison carried out in Section 6. The failure of the archimedean theory here is a relocation, not a degeneracy.

It is worth contrasting the situation with the rational base $3/2$. For the minimal word of that base, Stephan [12] obtains transcendence criteria, but the irrationality of the associated constant remains open; there the natural place is archimedean. Here it is the 2-adic place at which the argument closes, and the reason is the isometry of Proposition 2.2, which converts the approximation depth into a longest common prefix and so into the combinatorics of Sturmian words [9].

9. OPEN PROBLEMS

- (1) *All irrational slopes.* Is $\Phi(1c_\gamma)$ irrational for every irrational $\gamma \in (0, 1)$? The present proof needs exactly two inputs at the convergents of γ : a depth law at the convergents approaching from the side of the word, and a height bound of the shape $c_{w_n} \leq q_n \max(2^{q_n}, 3^{p_n})$ for the mechanical words. Lemma 4.2 holds verbatim for every slope; the delicate point is the depth law, and the two regimes $\gamma > \beta$ and $\gamma < \beta$ behave differently because the comparison in Theorem 6.1 weighs 2^{q_n} against 3^{p_n} , whose ratio changes side at $\gamma = \beta$.
- (2) *General intercepts.* The same for the mechanical words of slope β and intercept $\rho \neq 0$.
- (3) *An irrationality measure.* Is there $\mu < \infty$ and $C > 0$ with $|\Phi(1c_\beta) - y|_2 \geq C|y|^{-\mu}$ for all integers $y \geq 1$? By [5, Theorem 5.2] any finite μ would give an exponential lower bound on the least positive integers realising the prefixes of this word.
- (4) *Transcendence.* Is $\Phi(1c_\beta)$ transcendental over $\mathbb{Q}$? The archimedean analogue is known for the Hecke–Mahler values strictly inside the domain [3]; here one would want a 2-adic Mahler method along the continued-fraction chain.

None of these, and not Theorem 6.1 either, bears on divergent orbits of positive integers, on the Periodicity Conjecture in general, or on the Collatz conjecture.

APPENDIX A. VERIFICATION

The script `scripts/verify.py` accompanying this paper carries out the computations below in exact integer and 2-adic arithmetic; no floating-point value enters any decision. Each command is named with the claim it checks.

- **convergents** — Convention 5.1: the continued fraction of β to 368 rigorously determined partial quotients, the convergents p_n/q_n for $n \leq 14$, and the side of each, decided by the exact integer comparison 3^{p_n} against 2^{q_n} .
- **lcp** — Theorem 5.3 computed as a longest common prefix of the two 0/1 words.
- **depths** — Theorem 5.3 computed instead 2-adically, as v_2 of the difference of $\Phi(1c_\beta)$ and c_n/δ_n modulo a power of 2 exceeding the predicted depth. This is independent of the previous item, and agreeing values confirm Proposition 2.2 as well.
- **heights** — Lemma 4.2 for every mechanical word of slope p/q with $q \leq 120$, and the closed form $c_w = 2^a(3^a - 2^a)$ of Remark 4.3.
- **controls** — the two rational controls described below.
- **shadow** — the explicit height of Corollary 6.2.

Depths. Theorem 5.3 was verified at every level $3 \leq n \leq 14$, that is, out to the convergent 190537/301994. Each entry was computed twice and independently: as a longest common prefix of the two 0/1 words, and 2-adically as $v_2(\Phi(1c_\beta) - c_n/\delta_n)$ modulo 2^{L+40} , where L is the predicted depth; the deepest modulus used is 2^{478284} . All twenty-four values agree with the theorem.

n	p_n/q_n	predicted depth	computed
3	2/3	$q_3 + q_4 - 1 = 10$	10
4	5/8	$q_4 - 1 = 7$	7
5	12/19	$q_5 + q_6 - 1 = 83$	83
6	41/65	$q_6 - 1 = 64$	64
7	53/84	$q_7 + q_8 - 1 = 568$	568
8	306/485	$q_8 - 1 = 484$	484
9	665/1054	$q_9 + q_{10} - 1 = 25780$	25780
10	15601/24727	$q_{10} - 1 = 24726$	24726
11	31867/50508	$q_{11} + q_{12} - 1 = 176250$	176250
12	79335/125743	$q_{12} - 1 = 125742$	125742
13	111202/176251	$q_{13} + q_{14} - 1 = 478244$	478244
14	190537/301994	$q_{14} - 1 = 301993$	301993

Heights. Lemma 4.2 holds with no violation over every mechanical word $w_{p,q}$ with $q \leq 120$; the largest observed value of $c_w/(q \max(2^q, 3^p))$ is $1/3$, attained at $p/q = 1/1$.

Controls. A positive verification of a depth law is only meaningful if the same computation fails to produce a contradiction when run against a rational target, since the conclusion of Theorem 6.1 must not be derivable there. Two controls are computed. Taking as target the rational 2-adic integer $\Phi(w_3^\infty) = -5$, the depths $v_2(-5 - \Phi(w_n^\infty))$ do not grow: they freeze at $10 = q_3 + q_4 - 1$. Taking instead $\Phi(w_5^\infty)$, they freeze at $83 = q_5 + q_6 - 1$. In each case the agreement depth is bounded while the archimedean height of the approximants grows linearly, so the comparison of Section 6 yields nothing — exactly as it must not, these targets being rational.

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